

LAB 7

Introduction to Object-Oriented Programming

{OOP}

Polymorphism, abstraction

Lab07 Exercises

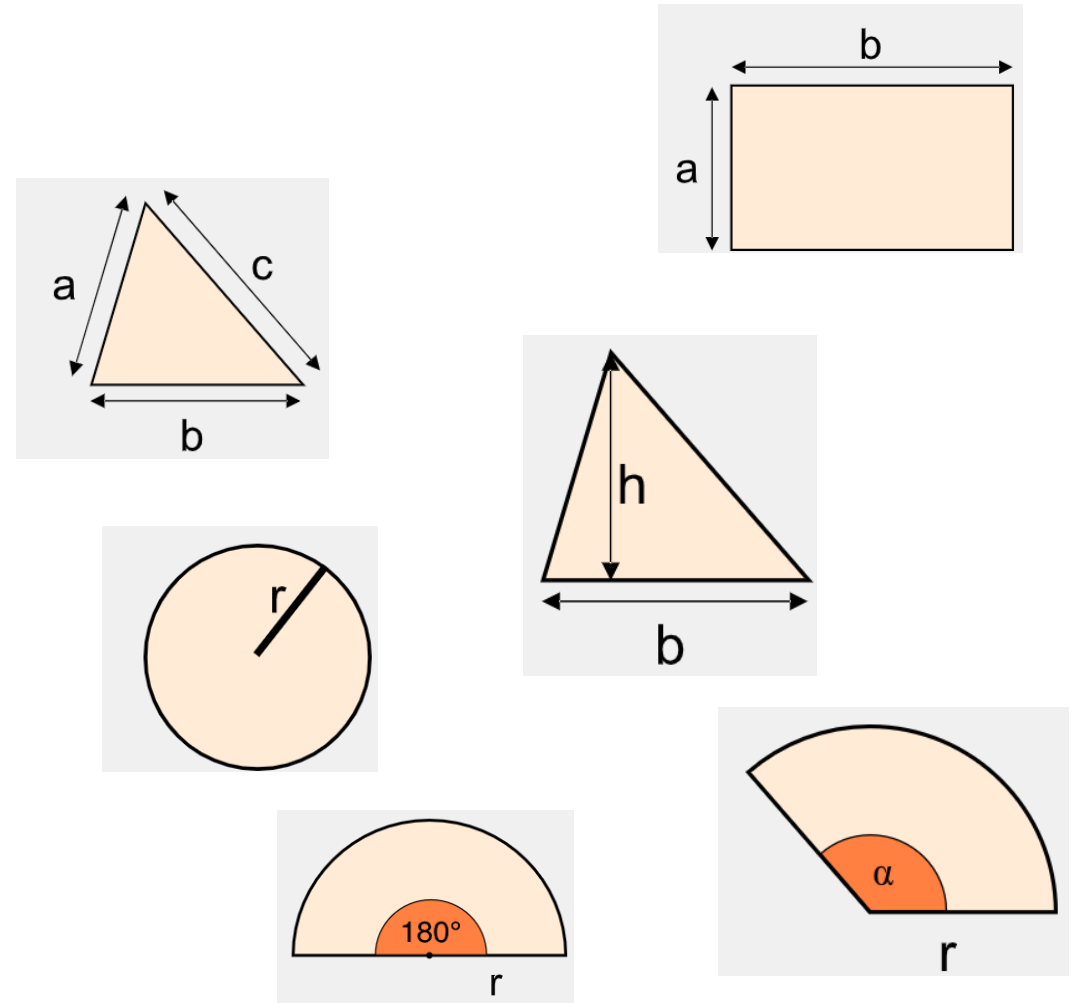
❖ Lab07.1. Implementing abstract class “ShapeArea”

Write a program to create a class “GeometryArea” and inherit from following “Abstract ShapeArea”
After all, override all functions and implement its own body statement to calculate accordingly.

```
abstract class ShapeArea {  
    abstract double calculateRectangleArea(double width, double height);  
    abstract double calculateCircleArea(double radius);  
    abstract double calculateTraingleArea(double a, double b, double c);  
    abstract double calculateTraingleArea(double base, double height);  
    abstract double calculateSemicircleArea(double radius);  
    abstract double calculateSectorArea(double radius, double angle);  
}
```

Console:

```
==== Shape Area Calculation ====  
1. Rectangle  
2. Circle  
3. Triangle (three sides)  
4. Triangle (base and height)  
5. Semi circle  
6. Sector  
7. Quit  
Choose an opt: 1
```



Lab07 Exercises

❖ Lab07.2. Number Randomizer

Write a program to create a class and inherit from abstract class “NumberRandomizer” for randomizing numbers. Implement the following function:

- **randomizeInBetween:** return a single randomized number in between two numbers
- **randomizeUniqueInBetween:** return array of randomized unique numbers in between two numbers

```
abstract class NumberRandomizer {  
    abstract int randomizeInBetween(int min, int max);  
    abstract int[] randomizeUniqueInBetween(int min, int max, int amount);  
}
```

Console:

```
==== Number Randomizer ===  
1. In between  
2. Unique in between  
3. Quit  
Choose an opt: 1
```

Lab07 Exercises

❖ Lab07.3. Text Randomizer

Write a program to create a class and inherit from abstract class “TextRadomizer” for randomizing character or string. Implement the following function:

- **randomizeACharacter**: return a single randomized character in alphabet
- **radomizeAString**: return a randomized string
- **radomizeUniqueStrings**: return an array of randomized string

```
abstract class TextRandomizer {  
    abstract char randomizeACharacter();  
    abstract String randomizeAString(int length);  
    abstract String[] randomizeStrings(int length, int amount);  
}
```

Console:

```
==== Text Randomizer ===  
1. A Character  
2. A String  
3. Unique Strings  
Choose an opt: 1
```

Lab07 Exercises

❖ Lab07.4. Nested Menu in Swing UI

Implement an application in Java to display and navigate Menu about “Setting”

*** Using **JPanel** to control different screen

Setting

1. General

2. Wi-Fi

3. Bluetooth

4. Mobile Data

5. Hotspot

6. Notification

Quit

Setting > General

1. About

2. Software update

3. Storage

Back

Setting > General > About

NameiPhone

ModeliXs

Version18.5

Back

Lab07 Exercises

❖ Lab07.5. Student management in JTable

Implement an application in Java to input a student information to a table.

*** Using Jtable and BorderLayout

Student Management

First name

Last name

Age

Major

Add new

First name	Last name	Age	Major
AA	Tola	21	IT
AA	Makara	19	IT
AA	Tola	21	IT
AA	Makara	19	IT
AA	Tola	21	IT
AA	Makara	19	IT
AA	Tola	21	IT
AA	Makara	19	IT

Create Jtable with header and row data

```
// Initial table header
DefaultTableModel header = new DefaultTableModel();
header.addColumn("First name");
header.addColumn("Last name");
header.addColumn("Age");
header.addColumn("Major");

// Initializing the JTable
JTable j = new JTable(header);
DefaultTableModel model = (DefaultTableModel) j.getModel();
model.addRow(new Object[]{ "AA", "Tola", "21", "IT" });
model.addRow(new Object[]{ "AA", "Makara", "19", "IT" });
```

Good luck 🤞

👉 How to submit my Lab works???

👉 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

Note: Don't zip the files for submission